B.Sc. (Hons.) Agriculture (First Semester) Examination, 2021-22 · Paper XI · 639(A)

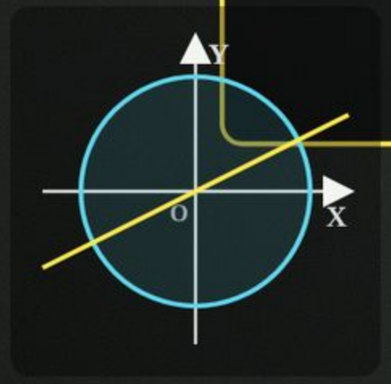
Elementary Mathematics

(प्रारंभिक गणित)

MAT-101 · Credit 2 (1+1)

Model Answer Paper (PYQ 2021-22 - all questions solved)

आदर्श उत्तर पत्र - पिछले वर्ष के सभी प्रश्न हल सहित, स्टेप-बाय-स्टेप



$\int f(x)\,dx$

dy/dx

$|A|$

Time	Maximum Marks	Sections
2 Hours	70	A: any 5 × 5 = 25 · B: any 3 × 15 = 45

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SECTION A

Short Answer Questions

लघु उत्तरीय प्रश्न

Attempt any five questions (all 8 solved here). Each carries 5 marks; answer within 50 words.

कोई पाँच प्रश्न कीजिए (यहाँ सभी 8 हल)। प्रत्येक 5 अंक; 50 शब्दों में।

5 × 5 = 25 Marks

A.1 **Find the coordinates of the point which divides internally the join of the points (8, 9) and (-7, 4) in the ratio 2 : 3.** **5 Marks**

(8, 9) और (-7, 4) को मिलाने वाली रेखा को 2 : 3 में अंतः विभाजित करने वाले बिंदु के निर्देशांक ज्ञात कीजिए।

Solution (हल)

Section formula (internal): अंतः विभाजन सूत्र

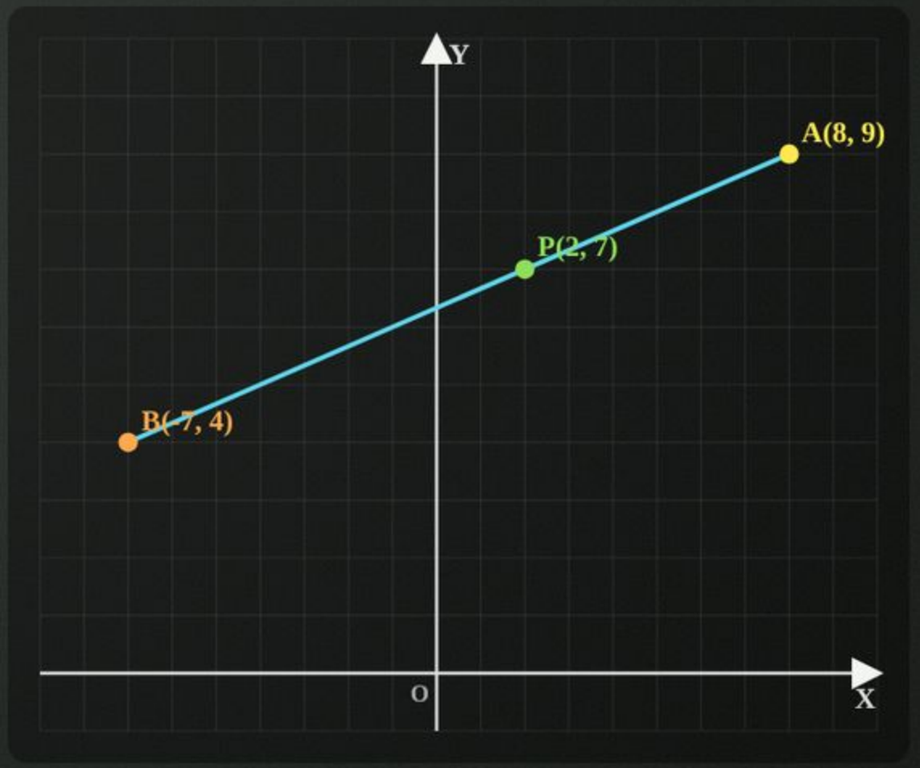
$$P = \left(\frac{m x_2 + n x_1}{m + n} , \frac{m y_2 + n y_1}{m + n} \right)$$

Here $(x_1, y_1) = (8, 9)$, $(x_2, y_2) = (-7, 4)$, $m : n = 2 : 3$ दिया है

$$x = \frac{2(-7) + 3(8)}{2 + 3} = \frac{-14 + 24}{5} = 2$$

$$y = \frac{2(4) + 3(9)}{2 + 3} = \frac{8 + 27}{5} = 7$$

Required point = **P (2, 7)**



P divides AB internally in 2 : 3

P, AB को 2 : 3 में अंतः विभाजित करता है

A.2 **Find the area of the quadrilateral whose vertices are A (-4, 5), B (0, 7), C (5, -5) and D (-4, -2).** **5 Marks**

उस चतुर्भुज का क्षेत्रफल ज्ञात कीजिए जिसके शीर्ष A (-4, 5), B (0, 7), C (5, -5) और D (-4, -2) हैं।

Solution (हल)

Area = ½ | Σ ($x_i\, y_{i+1} - x_{i+1}\, y_i$) | (vertices in order A, B, C, D) शीर्ष क्रम से लें

$x_A y_B - x_B y_A = (-4)(7) - (0)(5) = -28$

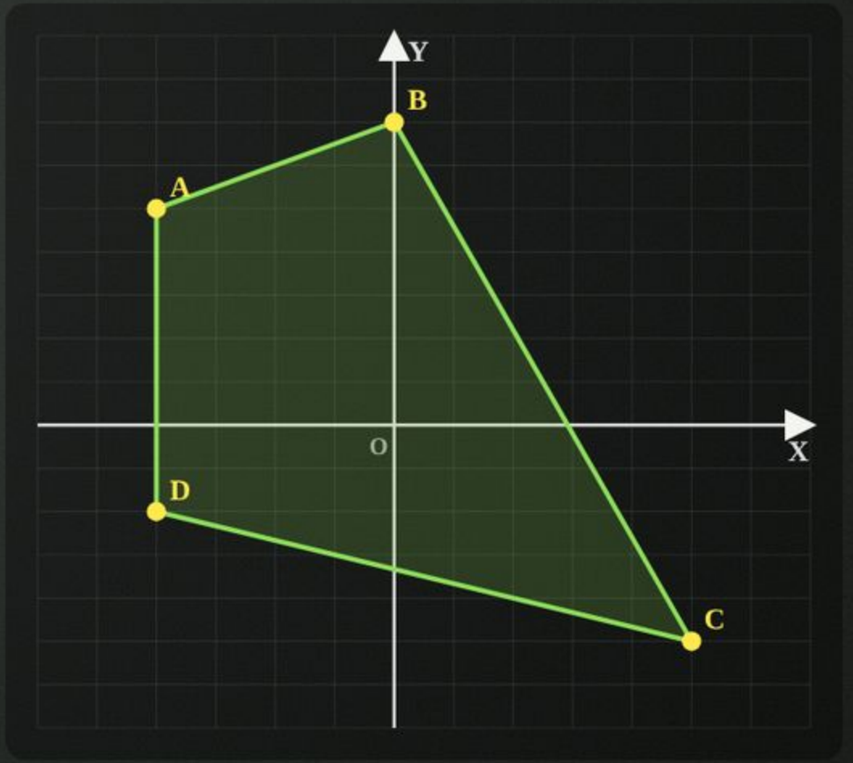
$x_B y_C - x_C y_B = (0)(-5) - (5)(7) = -35$

$x_C y_D - x_D y_C = (5)(-2) - (-4)(-5) = -30$

$x_D y_A - x_A y_D = (-4)(5) - (-4)(-2) = -28$

Sum = -28 - 35 - 30 - 28 = -121

Area = ½ × | -121 | = **60.5 sq. units**



Quadrilateral ABCD

चतुर्भुज ABCD

A.3 **Show that the straight lines $27x - 18y + 25 = 0$ and $2x + 3y + 7 = 0$ are perpendicular to each other.** **5 Marks**

दिखाइए कि सरल रेखाएँ $27x - 18y + 25 = 0$ और $2x + 3y + 7 = 0$ परस्पर लंबवत हैं।

Solution (हल)

For $ax + by + c = 0$, slope $m = \frac{-a}{b}$ ढाल का सूत्र

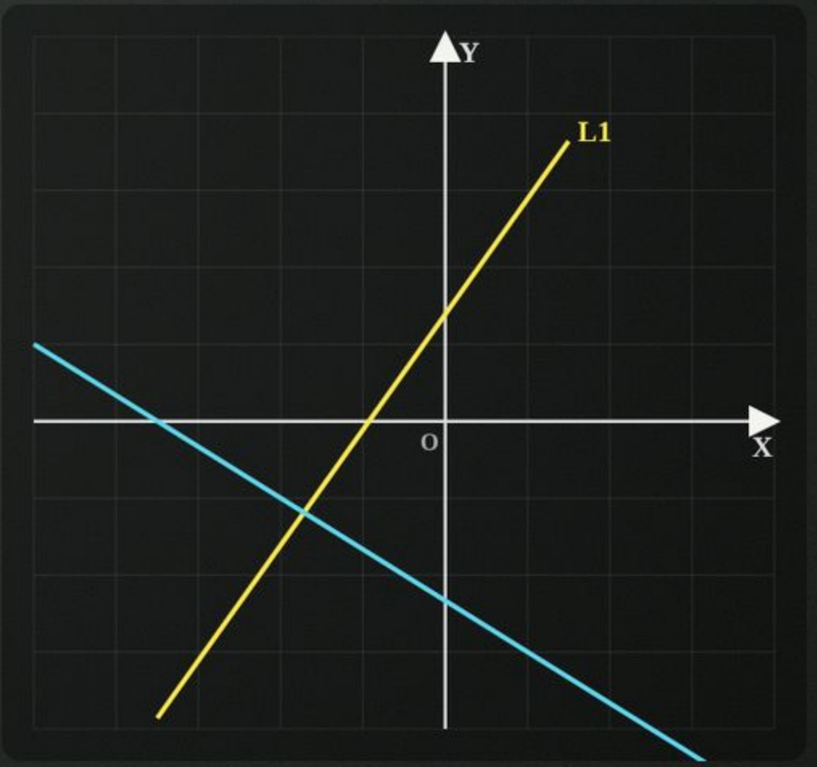
Line 1: $27x - 18y + 25 = 0 \Rightarrow m_1 = \frac{-27}{-18} = \frac{3}{2}$

Line 2: $2x + 3y + 7 = 0 \Rightarrow m_2 = \frac{-2}{3}$

$m_1 \times m_2 = \frac{3}{2} \times \frac{-2}{3} = -1$

Condition of perpendicular lines: $m_1\ m_2 = -1$ लंबवत होने की शर्त

Hence the two lines are perpendicular.



Lines meet at 90°

रेखाएँ 90° पर मिलती हैं

A.4 **Find the equations of tangent and normal to the circle $x^2 + y^2 = 25$ at (-3, 4).** **5 Marks**

वृत्त $x^2 + y^2 = 25$ के बिंदु (-3, 4) पर स्पर्श रेखा व अभिलंब के समीकरण ज्ञात कीजिए।

Solution (हल)

Tangent to $x^2 + y^2 = a^2$ at (x_1, y_1): $x\,x_1 + y\,y_1 = a^2$ स्पर्श रेखा का सूत्र

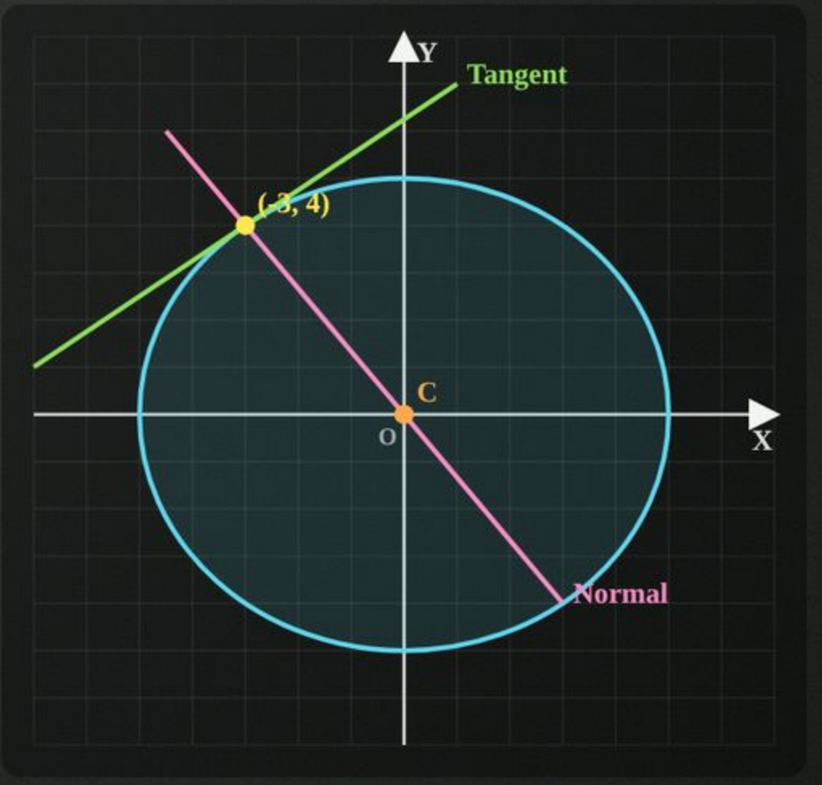
Tangent: $x(-3) + y(4) = 25 \Rightarrow$ **$3x - 4y + 25 = 0$**

Normal passes through the point (-3, 4) and the centre (0, 0). अभिलंब केंद्र से होकर जाता है

$$\text{Slope} = \frac{4 - 0}{-3 - 0} = \frac{-4}{3} \Rightarrow y = \frac{-4}{3}x$$

Normal: **$4x + 3y = 0$**

Check: $m_T \times m_N = (3/4)(-4/3) = -1$ ✓ जाँच



Tangent ⊥ radius at point of contact

स्पर्श बिंदु पर स्पर्श रेखा त्रिज्या पर लंब

A.5 **Prove that:** $\lim_{x \to 0} \frac{\log(1+x)}{x} = 1$ **5 Marks**

सिद्ध कीजिए कि $x \to 0$ पर $\log(1+x)/x$ की सीमा 1 है।

Solution (हल)

Method 1 (series): $\log(1+x) = x - x^2/2 + x^3/3 - \ldots$ for $|x| < 1$ लघुगणकीय श्रेणी

$$\frac{\log(1+x)}{x} = 1 - \frac{x}{2} + \frac{x^2}{3} - \ldots$$

$$\lim_{x \to 0} \left(1 - \frac{x}{2} + \frac{x^2}{3} - \ldots\right) = 1 - 0 + 0 = 1$$

Method 2: $\frac{1}{x}\log(1+x) = \log(1+x)^{1/x}$, and $\lim_{x \to 0}(1+x)^{1/x} = e$ दूसरी विधि

$\therefore$ limit $= \log e = 1$ **Hence proved: limit = 1**

A.6 **Prove that f(x) = |x| is continuous at x = 0.** **5 Marks**

सिद्ध कीजिए कि f(x) = |x|, x = 0 पर संतत है।

Solution (हल)

$|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$ मापांक फलन की परिभाषा

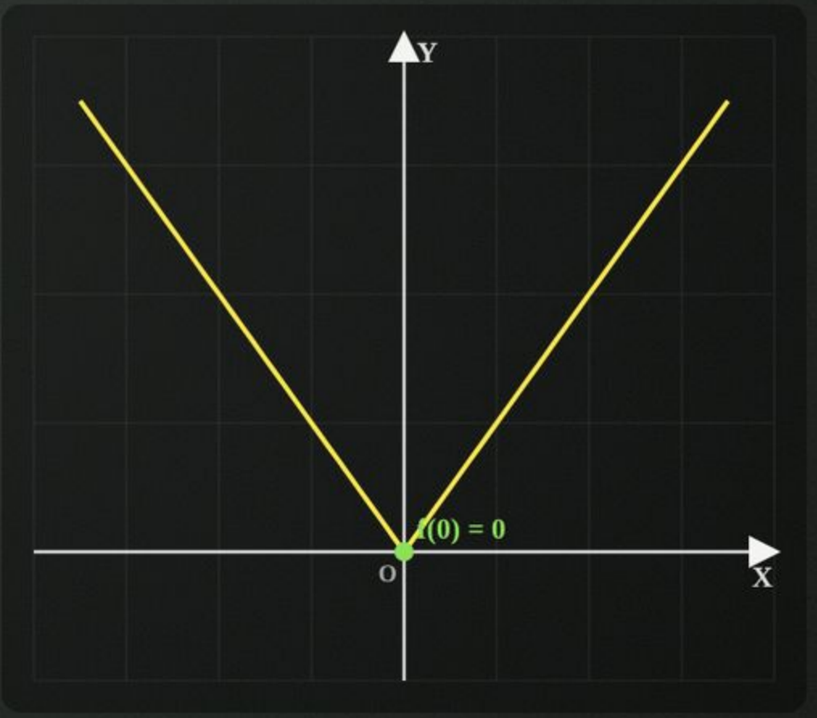
L.H.L. $= \lim_{h \to 0} f(0 - h) = \lim_{h \to 0} |-h| = \lim h = 0$ बायीं सीमा

R.H.L. $= \lim_{h \to 0} f(0 + h) = \lim_{h \to 0} |h| = 0$ दायीं सीमा

$f(0) = |0| = 0$ फलन का मान

L.H.L. = R.H.L. = f(0) = 0

∴ f(x) = |x| is continuous at x = 0



Graph of y = |x| has no break at 0

y = |x| का आलेख 0 पर टूटता नहीं

A.7 **Differentiate $e^{\tan x}$ with respect to x.** **5 Marks**

$e^{\tan x}$ का x के सापेक्ष अवकलन कीजिए।

Solution (हल)

Let $y = e^{\tan x}$. Put $u = \tan x$, so $y = e^u$ (chain rule) फलन का फलन नियम

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$\frac{dy}{du} = e^u = e^{\tan x}, \quad \frac{du}{dx} = \sec^2 x$$

$$\frac{dy}{dx} = \boxed{e^{\tan x} \cdot \sec^2 x}$$

Remember (याद रखें)

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} e^{f(x)} = e^{f(x)} \cdot f'(x)$$

A.8 **Find the maximum and minimum value of the polynomial $y = 3x^4 - 2x^3 - 6x^2 + 6x + 1$.** **5 Marks**

बहुपद $y = 3x^4 - 2x^3 - 6x^2 + 6x + 1$ के उच्चिष्ठ व निम्निष्ठ मान ज्ञात कीजिए।

Solution (हल)

$\frac{dy}{dx} = 12x^3 - 6x^2 - 12x + 6 = 6(2x - 1)(x^2 - 1)$

For extreme values $\frac{dy}{dx} = 0 \Rightarrow x = ½, 1, -1$ चरम मानों हेतु

$\frac{d^2y}{dx^2} = 36x^2 - 12x - 12$

At $x = ½$: $y'' = 9 - 6 - 12 = -9 < 0 \Rightarrow$ maximum, $y = 39/16$

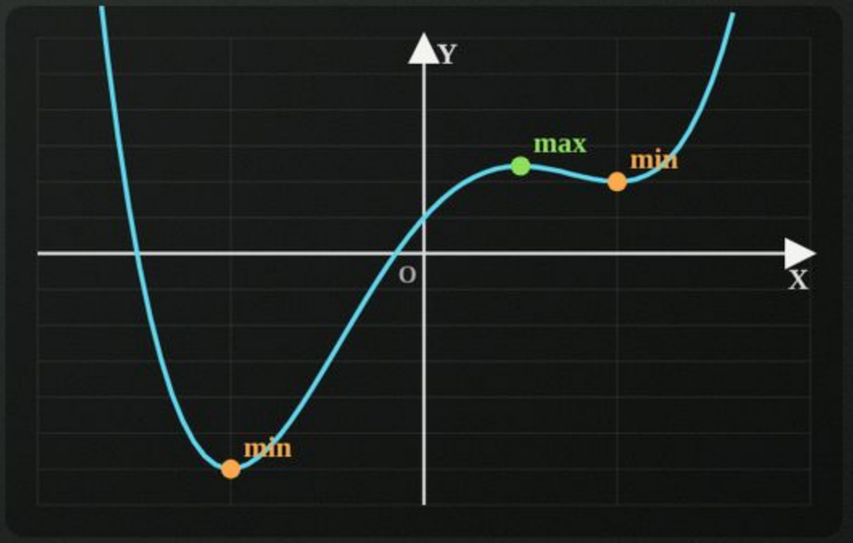
At $x = 1$: $y'' = 12 > 0 \Rightarrow$ minimum, $y = 3 - 2 - 6 + 6 + 1 = 2$

At $x = -1$: $y'' = 36 > 0 \Rightarrow$ minimum, $y = 3 + 2 - 6 - 6 + 1 = -6$

Answer (उत्तर)

Maximum value = 39/16 (at $x = ½$)

Minimum values = 2 ($x = 1$) and -6 ($x = -1$)



SECTION B

Long Answer Questions

दीर्घ उत्तरीय प्रश्न

Attempt any three questions (all 5 solved here). Each question carries 15 marks.

कोई तीन प्रश्न कीजिए (यहाँ सभी 5 हल)। प्रत्येक प्रश्न 15 अंक का है।

3 × 15 = 45 Marks

B.1 **Evaluate: (i) $\int x\sqrt{(1+x)}\,dx$ (ii) $\int_0^1 (1+x^2)^3\,dx$** **15 Marks**

मान ज्ञात कीजिए: (i) $\int x\sqrt{(1+x)}\,dx$ (ii) 0 से 1 तक $\int (1+x^2)^3\,dx$

Part (i) (भाग (i))

(i) Put $1 + x = t \Rightarrow dx = dt$, $x = t - 1$ प्रतिस्थापन विधि

$$I = \int (t-1)\,t^{1/2}\,dt = \int (t^{3/2} - t^{1/2})\,dt$$

$$= \frac{t^{5/2}}{5/2} - \frac{t^{3/2}}{3/2} + C$$

$$= \frac{2}{5}\,t^{5/2} - \frac{2}{3}\,t^{3/2} + C$$

$I = (2/5)(1+x)^{5/2} - (2/3)(1+x)^{3/2} + C$

Part (ii) (भाग (ii))

(ii) Expand: $(1+x^2)^3 = 1 + 3x^2 + 3x^4 + x^6$ द्विपद प्रसार

$$I = \left[\, x + x^3 + \frac{3x^5}{5} + \frac{x^7}{7} \,\right]_0^1$$

$$= 1 + 1 + \frac{3}{5} + \frac{1}{7} - 0$$

$$= \frac{70 + 21 + 5}{35}$$

$I = 96/35 \approx 2.743$

Tip: write "+ C" in indefinite integrals; in definite integrals put upper limit minus lower limit.

अनिश्चित समाकल में + C लिखें; निश्चित समाकल में ऊपरी सीमा का मान घटा निचली सीमा का मान।

B.2 **Using integration, find the areas of the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ and the circle $x^2 + y^2 = 9$.** **15 Marks**

समाकलन द्वारा दीर्घवृत्त $x^2/4 + y^2/9 = 1$ तथा वृत्त $x^2 + y^2 = 9$ के क्षेत्रफल ज्ञात कीजिए।

Area of ellipse (दीर्घवृत्त का क्षेत्रफल)

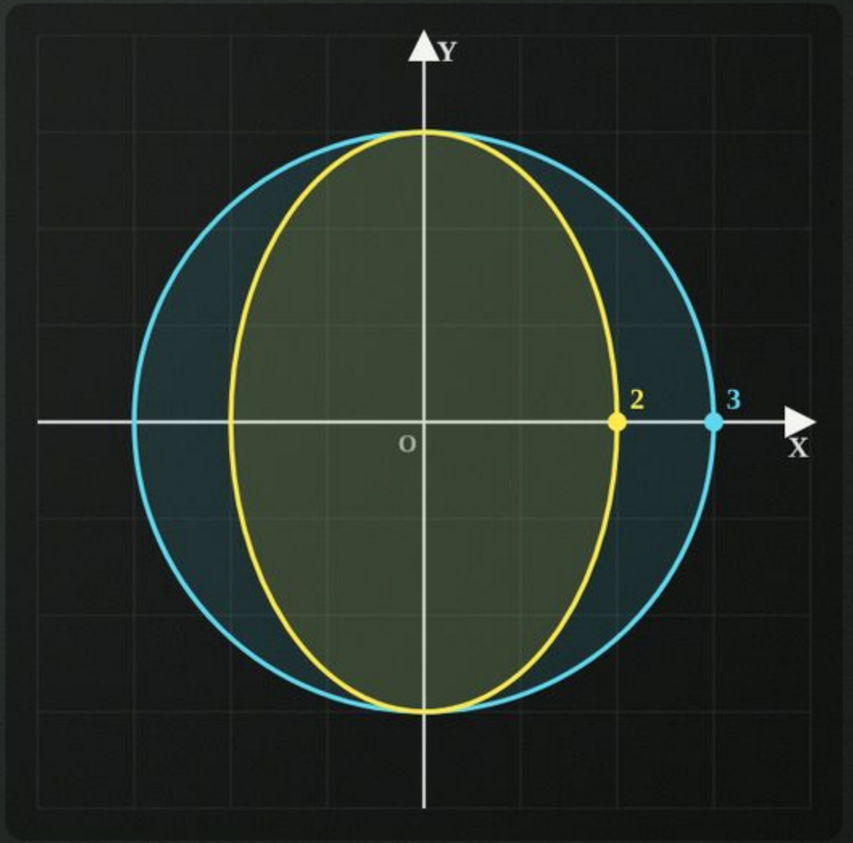
Ellipse: $y = (3/2)\sqrt{(4 - x^2)}$; by symmetry area = 4 × area in 1st quadrant सममिति से चार गुना

$$A_1 = 4 \int_0^2 (3/2) \sqrt{(4 - x^2)}\, dx = 6 \int_0^2 \sqrt{(4 - x^2)}\, dx$$

Using $\int \sqrt{(a^2 - x^2)}\, dx = \frac{x}{2} \sqrt{(a^2 - x^2)} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$

$$A_1 = 6 [\, 0 + 2 \sin^{-1}(1) - 0 \,] = 6 \times 2 \times \frac{\pi}{2} = \boxed{6\pi \text{ sq. units}}$$

Check: area of ellipse = $\pi ab = \pi \times 2 \times 3 = 6\pi$ ✓ जाँच



Yellow: ellipse (a = 2, b = 3); blue: circle (r = 3)

पीला: दीर्घवृत्त; नीला: वृत्त

B.2 Area of the circle $x^2 + y^2 = 9$

वृत्त $x^2 + y^2 = 9$ का क्षेत्रफल

contd.

Area of circle (वृत्त का क्षेत्रफल)

$y = \sqrt{(9 - x^2)}$; area = 4 × area in first quadrant प्रथम चतुर्थांश का चार गुना

$$A_2 = 4 \int_0^3 \sqrt{(9 - x^2)}\, dx$$

$$= 4 \left[\frac{x}{2} \sqrt{(9 - x^2)} + \frac{9}{2} \sin^{-1} \frac{x}{3} \right]_0^3$$

$$= 4 \left[0 + \frac{9}{2} \times \frac{\pi}{2} - 0 \right] = 4 \times \frac{9\pi}{4}$$

$A_2 =$ **9π sq. units** (check: $\pi r^2 = 9\pi$ ✓)

Final answer (अंतिम उत्तर)

Ellipse = $6\pi \approx 18.85$ sq. units

Circle = $9\pi \approx 28.27$ sq. units

Area between them = 3π

B.3 **Using properties of determinants, prove that:** $\begin{vmatrix} x^2 & y^2 & z^2 \\ yz & zx & xy \\ x & y & z \end{vmatrix} = (x - y)(y - z)(z - x)(xy + yz + zx)$

15 Marks

सारणिक के गुणधर्मों से सिद्ध कीजिए।

Solution (हल)

Multiply C_1 by x, C_2 by y, C_3 by z and divide by xyz: स्तंभों को x, y, z से गुणा

$$\Delta = \frac{1}{xyz}\begin{vmatrix} x^3 & y^3 & z^3 \\ xyz & xyz & xyz \\ x^2 & y^2 & z^2 \end{vmatrix} = \begin{vmatrix} x^3 & y^3 & z^3 \\ 1 & 1 & 1 \\ x^2 & y^2 & z^2 \end{vmatrix}$$ (xyz taken out of R_2)

$C_1 \to C_1 - C_2, C_2 \to C_2 - C_3$: $\Delta = \begin{vmatrix} x^3 - y^3 & y^3 - z^3 & z^3 \\ 0 & 0 & 1 \\ x^2 - y^2 & y^2 - z^2 & z^2 \end{vmatrix}$

B.3 Expansion and final result
प्रसार व अंतिम परिणाम

contd.

Solution (हल)

Expand along R_2 (only $a_{23} = 1$ is non-zero; its sign is -): R_2 के अनुदिश प्रसार

$$\Delta = - \begin{vmatrix} x^3 - y^3 & y^3 - z^3 \\ x^2 - y^2 & y^2 - z^2 \end{vmatrix} = - (x - y)(y - z) \begin{vmatrix} x^2 + xy + y^2 & y^2 + yz + z^2 \\ x + y & y + z \end{vmatrix}$$

$$= - (x - y)(y - z) [(x^2 + xy + y^2)(y + z) - (y^2 + yz + z^2)(x + y)]$$

$$= - (x - y)(y - z) [x^2y + x^2z - xz^2 - yz^2] = - (x - y)(y - z)(x - z)(xy + yz + zx)$$

$$\therefore \Delta = \boxed{(x - y)(y - z)(z - x)(xy + yz + zx)}$$ Hence proved.

Properties used: a common factor of a row/column can be taken out; $C_i \to C_i - C_j$ does not change the value; expansion along a row.

प्रयुक्त गुणधर्म: पंक्ति/स्तंभ से उभयनिष्ठ गुणनखंड बाहर; $C_i - C_j$ से मान नहीं बदलता; पंक्ति के अनुदिश प्रसार।

B.4 If $A = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}$, then verify that $(AB)^{-1} = B^{-1}A^{-1}$.

15 Marks

यदि A व B दिए हैं, तो सत्यापित कीजिए कि $(AB)^{-1} = B^{-1}A^{-1}$

Step 1: $(AB)^{-1}$ (चरण 1)

$$AB = \begin{pmatrix} 2\cdot0 + 0\cdot2 & 2\cdot1 + 0\cdot4 \\ 3\cdot0 + 1\cdot2 & 3\cdot1 + 1\cdot4 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 2 & 7 \end{pmatrix}$$

$$|AB| = 0 \times 7 - 2 \times 2 = -4 \neq 0$$

$$(AB)^{-1} = \frac{\text{adj}\,(AB)}{|AB|} = \frac{1}{-4}\begin{pmatrix} 7 & -2 \\ -2 & 0 \end{pmatrix}$$

$$(AB)^{-1} = \begin{pmatrix} -7/4 & 1/2 \\ 1/2 & 0 \end{pmatrix} \quad \text{... (1)}$$

Step 2: A^{-1} and B^{-1} (चरण 2)

$$|A| = 2, \quad A^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 1/2 & 0 \\ -3/2 & 1 \end{pmatrix}$$

$$|B| = -2, \quad B^{-1} = \frac{1}{-2}\begin{pmatrix} 4 & -1 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 1/2 \\ 1 & 0 \end{pmatrix}$$

B.4 Step 3: $B^{-1}A^{-1}$ and verification

चरण 3: $B^{-1}A^{-1}$ व सत्यापन

contd.

Solution (हल)

$$B^{-1}A^{-1} = \begin{pmatrix} -2 & 1/2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1/2 & 0 \\ -3/2 & 1 \end{pmatrix} = \begin{pmatrix} -1 - 3/4 & 0 + 1/2 \\ 1/2 + 0 & 0 + 0 \end{pmatrix} = \begin{pmatrix} -7/4 & 1/2 \\ 1/2 & 0 \end{pmatrix} \quad \text{... (2)}$$

From (1) and (2): (1) और (2) से

$(AB)^{-1} = B^{-1}A^{-1}$ Hence verified.

Formula used (प्रयुक्त सूत्र)

For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$: $M^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (swap diagonal, change sign of off-diagonal)

B.5 **(i) Find the equation of straight line which passes through the point (3, 4) and the sum of whose intercepts on the axes is 14. (ii) Find the equation of the circle of radius 5 units, whose centre lies on the x-axis and which passes through the point (2, 3).** **15 Marks**

(i) (3, 4) से होकर जाने वाली रेखा जिसके अंतःखंडों का योग 14 है। (ii) त्रिज्या 5, केंद्र x-अक्ष पर, (2, 3) से होकर जाने वाला वृत्त।

Part (i) (भाग (i))

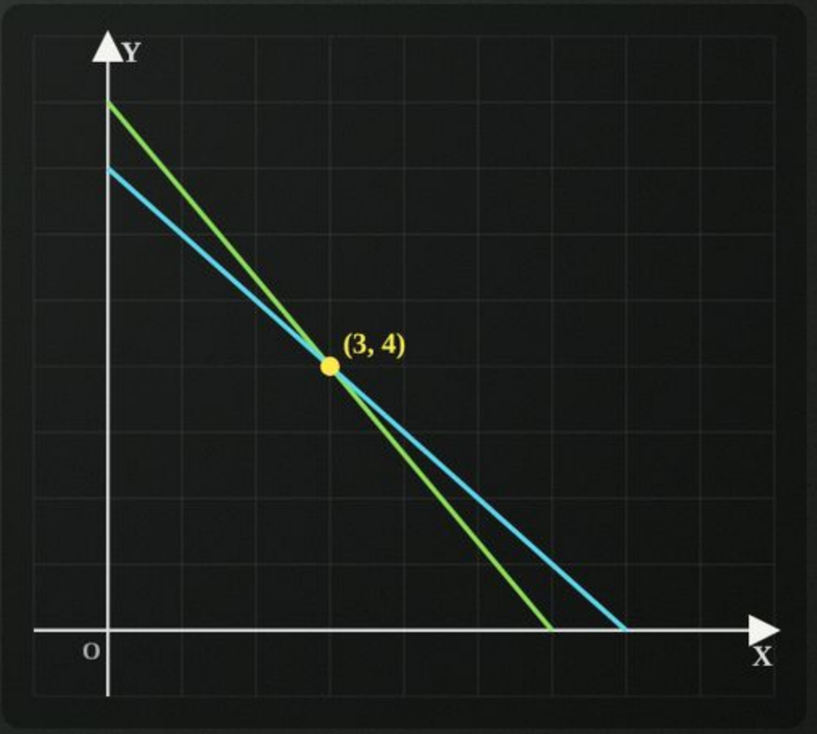
Intercept form: $x/a + y/b = 1$, with $a + b = 14 \Rightarrow b = 14 - a$ अंतःखंड रूप

Passes through (3, 4): $\frac{3}{a} + \frac{4}{14 - a} = 1$

$3(14 - a) + 4a = a(14 - a) \Rightarrow a^2 - 13a + 42 = 0 \Rightarrow (a - 6)(a - 7) = 0$

$a = 6, b = 8$: $\frac{x}{6} + \frac{y}{8} = 1 \Rightarrow$ **$4x + 3y = 24$**

$a = 7, b = 7$: $\frac{x}{7} + \frac{y}{7} = 1 \Rightarrow$ **$x + y = 7$**



Two lines satisfy the condition

दो रेखाएँ शर्त पूरी करती हैं

B.5 (ii) Circle of radius 5 with centre on x-axis through (2, 3)

(ii) वृत्त का समीकरण

contd.

Part (ii) (भाग (ii))

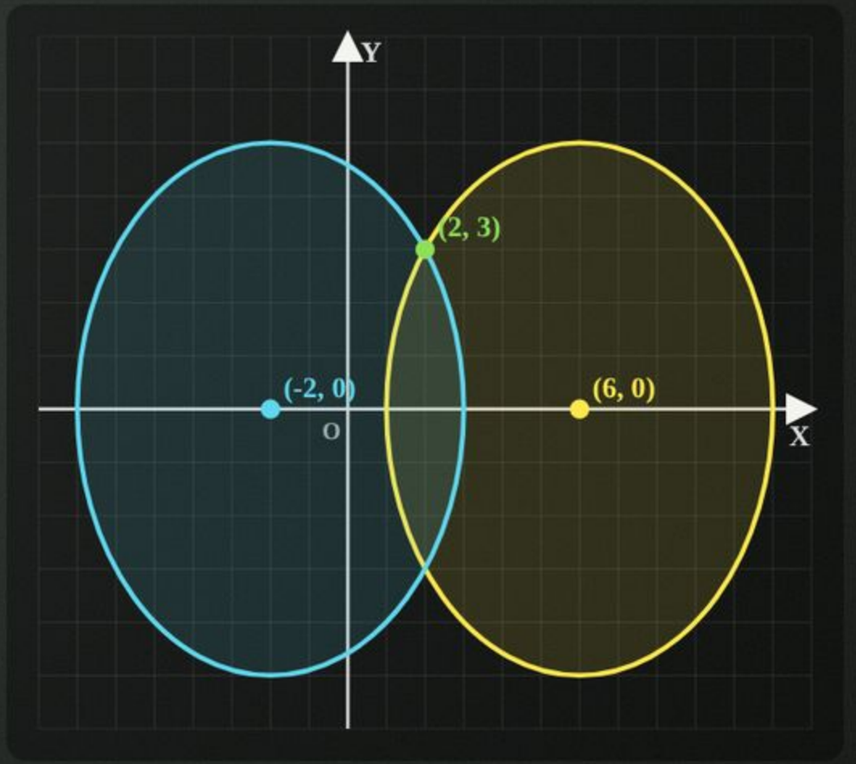
Centre on x-axis ⇒ centre (h, 0); equation $(x - h)^2 + y^2 = 25$ केंद्र x-अक्ष पर

Passes through (2, 3): $(2 - h)^2 + 9 = 25 \Rightarrow (2 - h)^2 = 16$

$2 - h = \pm 4 \Rightarrow h = -2$ or $h = 6$

$h = 6$: $(x - 6)^2 + y^2 = 25 \Rightarrow$ **$x^2 + y^2 - 12x + 11 = 0$**

$h = -2$: $(x + 2)^2 + y^2 = 25 \Rightarrow$ **$x^2 + y^2 + 4x - 21 = 0$**



Both circles pass through (2, 3)

दोनों वृत्त (2, 3) से होकर